

Source-grouped evaluation and temporal stacking

Curated STEAD input

2234 records / 2114 sources
E, N, Z; 100 Hz; 60 s



Source-disjoint partition

Train 1480 / validation 317 / test 317 sources
Test: 335 records; no source overlap



Temporal meta-learner

12 probability + 3 waveform channels
I-CNN; seeds 42, 43, 44
Final base/meta reuse outer validation



Base models and OOF

STA/LTA, AIC, CNN, TCN
CNN/TCN: eight input channels
Five source folds + inner validation



Fixed arrival decoder

P maximum; S search P + 0.1–30 s
Detected and uncertain accepted
Strict sensitivity retained



Frozen test evaluation

F1 at 50, 100, 200 ms; accepted-pick MAE
Full3C / Z-only and learner ablations
Three-seed means; source-cluster intervals